

1. Raw audio is high dimensional and doesn't show the frequency content. Converting into a spectrogram or MFCC representation compresses the signal into a compact time-frequency array. MFCCs use the Mel scale to match human hearing, which makes the patterns easier for a small TinyML model to learn and allows us to deploy it on our microcontroller.
2. The representative dataset is necessary for the model to correctly map float32 activations and weights onto the int8 range. The converter needs to know the actual numeric range of each layer's activation to ensure accurate quantization. This allows it to calibrate the scale and zero point using realistic inputs instead of an unnecessarily large range.
3. It pulls a batch of audio samples from a data split (training, testing) and applies background noise and random time shifts, then runs each clip through the feature extraction pipeline, then returns the resulting feature arrays and their labels.
4. `g_model` is a byte array that contains the full quantized TensorFlow Lite model, embedded into the Arduino firmware. `g_model_len` is the byte length of the array, which is needed because C arrays do not store their size like arrays in other languages do. Both must be copied into the source file because a mismatch in these values could cause parts of the model to not be loaded and output garbage values, or cause an error and not load at all.
5. I used "hello" to test the unknown. It had a harder time detecting "no" than detecting "yes" for me. It would label "no" as unknown very frequently.

```
Output  Serial Monitor X
Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM4')

Heard unknown (201) @56848ms
Heard unknown (201) @60176ms
Heard yes (219) @62752ms
Heard yes (210) @65056ms
Heard no (201) @67264ms
Heard unknown (211) @69104ms
Heard unknown (201) @71136ms
Heard unknown (202) @75824ms
Heard yes (210) @78592ms
Heard unknown (204) @83200ms
Heard unknown (216) @84768ms
Heard unknown (209) @90384ms
Heard yes (218) @92784ms
Heard unknown (203) @103920ms
Heard unknown (201) @105488ms
```